

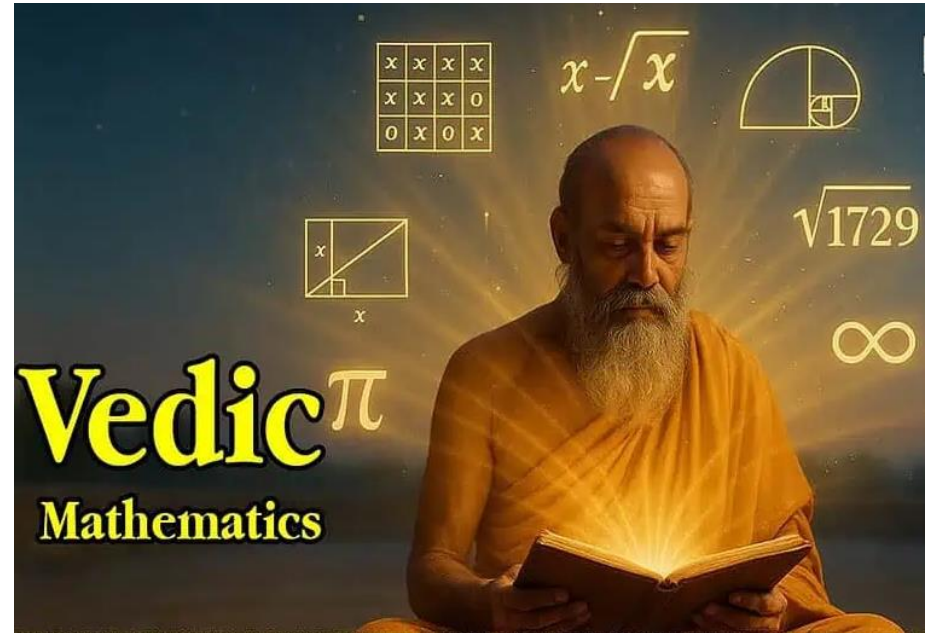
Chapter 5: Numbers in the Vedic Literature

The Mathematics of India by P. P. Divakaran



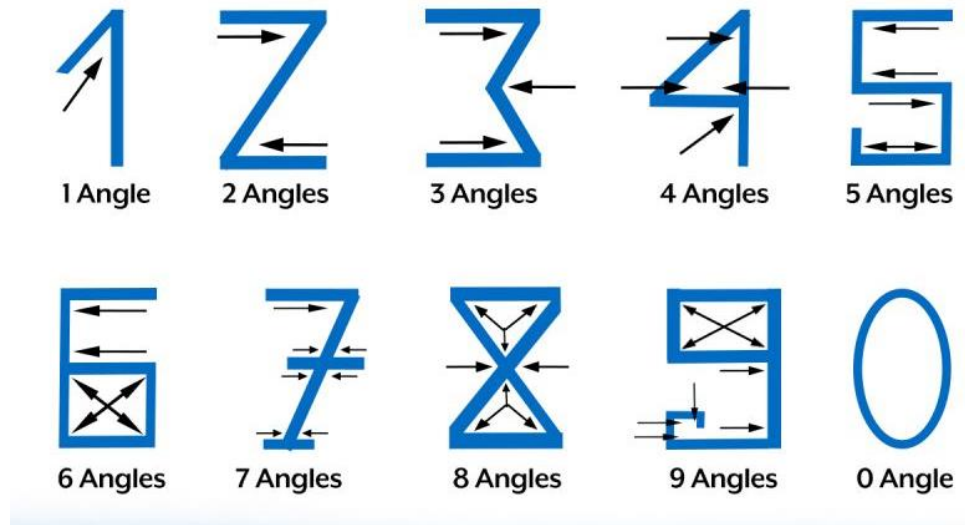
Introduction to Vedic Numbers

- Vedic literature contains some of the earliest references to numbers in India
- Numbers were deeply connected to ritual, cosmology, and philosophy
- Counting and enumeration were essential for Vedic sacrifices
- Large numbers appear frequently in hymns and ritual texts
- This chapter explores how numerical ideas developed in the Vedic age



Origins of Number Use

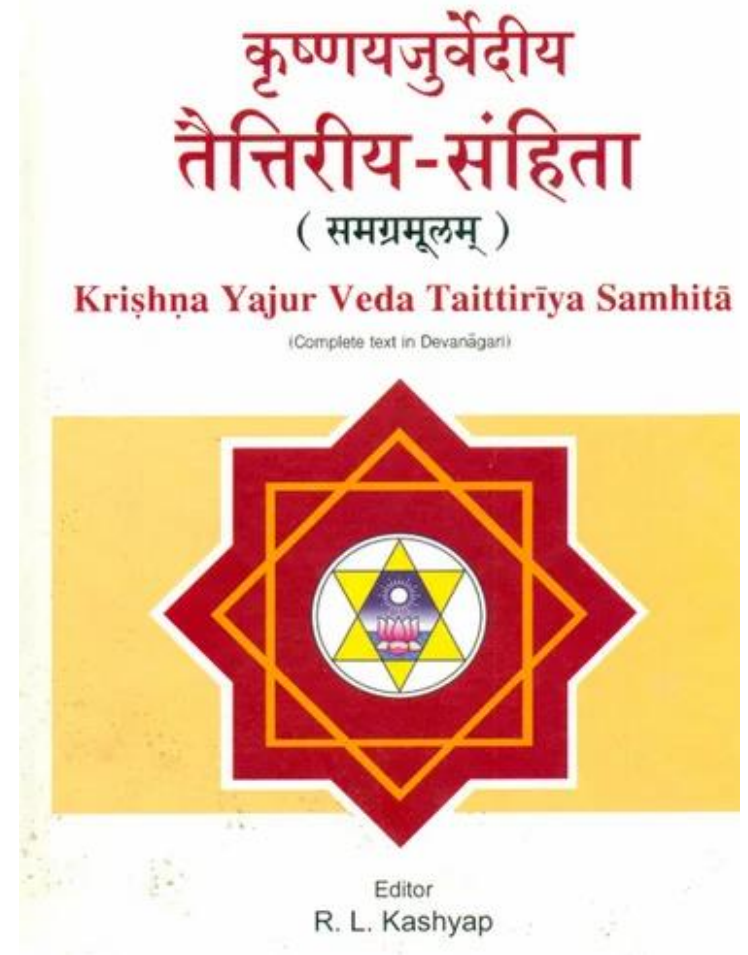
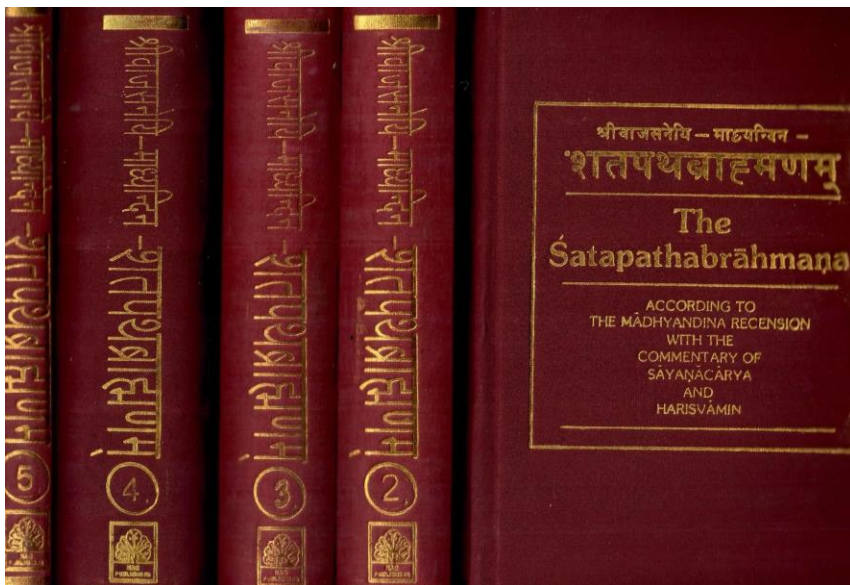
- Numbers first appear in the Rigveda in poetic and symbolic contexts
- They were used to describe quantities of offerings, ritual objects, altar construction and cosmology
- Enumeration helped structure hymns and ritual procedures



- Early number names reveal a strong decimal tendency
- These origins laid the foundation for later Indian arithmetic

Number Names in Early Texts

- Gujardesa (595 CE) : World's oldest written document that uses present day numerals 1, 2, 3.....
- Mohenjo-Daro writings (3000 – 6000 BCE) : Evidence of written numbers from 1 to 13 and written language in Brahmi script is found on the vessels



- Gwalior inscription (870 CE) : The earliest known written zero

Decimal Structure Emerging

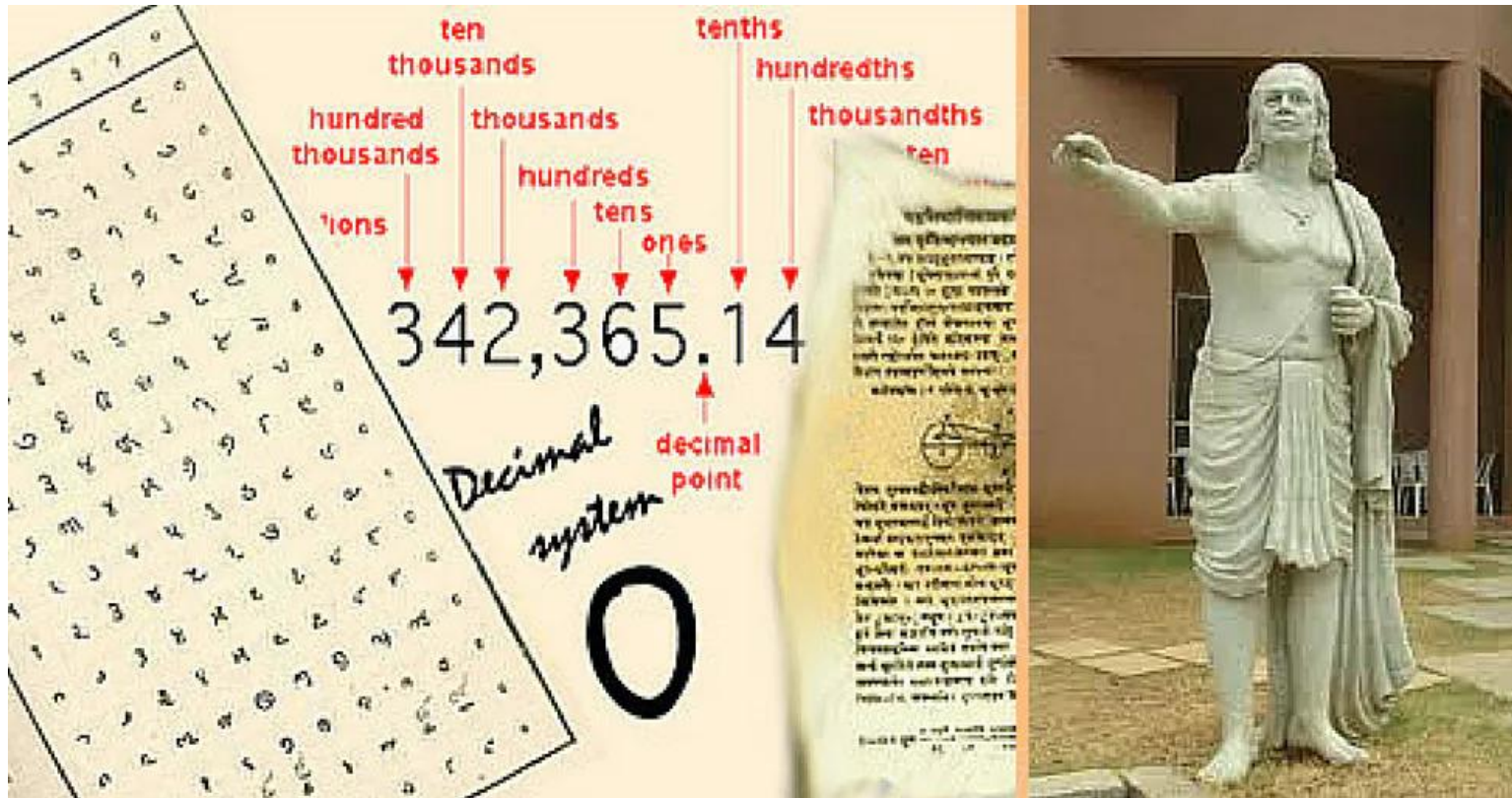


Figure :
Hindi
Arabic
numeral
system

- Vyasa's commentary on Patanjali's Yoga Sutras, known as the Yogabhāṣya (योगभाष्य) (4th or 5th century CE) beautifully explains the place value system
- “Just as a single stroke (symbol) can represent one, ten, hundred, or thousand depending on its place”

16 basic sutra of Vedic Mathematics

- 1) *Ekadhikina Purvena* – By one more than the previous One
- 2) *Nikhilam Navatashcaramam Dashatah* – All from 9 and last from 10
- 3) *Urdhva-tiryakbhyam* – Vertically and crosswise
- 4) *Paraavartya Yojayet* – Transpose and adjust
- 5) *Shunyam Saamyasamuccaye* – When the sum is same then sum is zero
- 6) *(Anurupye) Shunyamanyat* – If one is in ratio, the other is zero
- 7) *Sankalana-vyavakalanabhyam* – By addition and by subtraction
- 8) *Puranapuranaabyham* – By the completion non-completion

SIXTEEN SŪTRAS AND THEIR COROLLARIES

Sūtras	Sub-Sūtras or Corollaries
1. एकाधिकेन पूर्वेण <i>Ekādhikena Pūrveṇa</i> (also a corollary)	1. आनुरूप्येण <i>Ānurūpyeṇa</i>
2. निखिलं नवतश्चरमं दशतः <i>Nikhilaṁ Navataścaramaṁ Daśataḥ</i>	2. शिष्यते शेषसंज्ञः <i>Śiṣyate Śeṣasaṁjñāḥ</i>
3. ऊर्ध्वतिर्यग्भ्याम् <i>Ūrdhva-tiryagbhyām</i>	3. आद्यमाद्ये नान्त्यमन्त्येन <i>Ādyamādyenāntyamantyaena</i>
4. परावर्त्य योजयेत् <i>Parāvartya Yojayet</i>	4. केवलैः सप्तकं गुण्यात् <i>Kevalaiḥ Saptakaṁ Guṇyāt</i>
5. शून्यं साम्यसमुच्चये <i>Śūnyaṁ Sāmyasamuccaye</i>	5. वेष्टनम् <i>Veṣṭanam</i>
6. (आनुरूप्ये) शून्यमन्यत् <i>(Ānurūpye) Śūnyamanyat</i>	6. यावदूनं तावदूनम् <i>Yāvadūnaṁ Tāvadūnaṁ</i>
7. संकलनव्यवकलनाभ्याम् <i>Saṅkalana-vyavakalanābhyām</i> (also a corollary)	7. यावदूनं तावदूनीकृत्य वर्गं च योजयेत् <i>Yāvadunaṁ Tāvadūnikṛtya Vargaṅca Yojayet</i>
8. पूरणापूरणाभ्याम् <i>Pūraṇāpūraṇābhyām</i>	8. अन्त्ययोर्दशकेऽपि <i>Antyayordaśake 'pi</i>
9. चलनकलनाभ्याम् <i>Calana-kalanābhyām</i>	9. अन्त्ययोरेव <i>Antyayoreva</i>
10. यावदूनम् <i>Yāvadūnam</i>	10. समुच्चयगुणितः <i>Samuccayagunitaḥ</i>
11. व्यष्टिसमष्टिः <i>Vyaṣṭisamaṣṭiḥ</i>	11. लोपनस्थापनाभ्याम् <i>Lopanasthāpanābhyām</i>

16 basic sutra of Vedic Mathematics

- 9) *ChalanaKalanabyham* – Differences and similarities
- 10) *Yaavadunam* – Whatever the extent of its deficiency
- 11) *Vyashtisamanstih* – Part and Whole
- 12) *Shesanyankena Charamena* – The remainders by the last digit
- 13) *Sopaantyadvayamantyam* – The ultimate and twice the penultimate
- 14) *Ekanyunena Purvena* – By one less than the previous one
- 15) *Gunitasamuchyah* – The product of sum is equal to sum of the product
- 16) *Gunakasamuchyah* – Factors of the sum is equal to the sum of factors

12. शेषाण्यङ्केन चरमेण
Śeṣāṇyāṅkena Carameṇa
13. सोपान्त्यद्वयमन्त्यम्
Sopāntyadvayamantyam
14. एकन्यूनेन पूर्वेण
Ekanyūnena Pūrveṇa
15. गुणितसमुच्चयः
Guṇitasamuccayaḥ
16. गुणकसमुच्चयः
Guṇakasamuccayaḥ

12. विलोकनम्
Vilokanam
13. गुणितसमुच्चयः समुच्चयगुणितः
*Guṇitasamuccayaḥ
Samuccayaguṇitaḥ*

Large Numbers in the Vedas

- Texts mention numbers extending to extremely high magnitudes which shows that the concept of very large numbers fascinated Vedic thinkers
- These numbers often had ritual or symbolic meanings
- Rigveda mentions name of numbers 1 to 9 and also 10, 20, ..90

Reference : Taittirīya Samhitā (1200-900 BCE)

<i>ekam</i>	1
<i>daśa</i>	10
<i>śatam</i>	10^2
<i>sahasram</i>	10^3
<i>ayutam</i>	10^4
<i>niyutam</i>	10^5
<i>prayutam</i>	10^6
<i>kotiḥ</i>	10^7
<i>arbudam</i>	10^8
<i>vṛndam</i>	10^9
<i>kharvaḥ</i>	10^{10}
<i>nikharvaḥ</i>	10^{11}
<i>mahāpadmaḥ</i>	10^{12}
<i>śaṅkuḥ</i>	10^{13}
<i>vāridhiḥ</i>	10^{14}
<i>antyam</i>	10^{15}
<i>madhyam</i>	10^{16}
<i>parārdham</i>	10^{17}

Large Numbers in the Vedas

- In the first century CE, a famous Buddhist text called *Lalitavistara* records a conversation in which the mathematician Arjuna asks young Prince Gautama (the Bodhisattva) to recite large number names. These numbers often had ritual or symbolic meanings
- Gautama lists names of powers of 10, starting from *koti* (10^7) and going all the way up to *tallaksana* (10^{53})!
- The last number is unimaginably large—big enough to represent the number of grains of sand that could fill a heap the size of the entire universe
- Also, in the Jain text *Anuyogadvārasūtra* (1st century BCE), very large numbers such as *asankhyeya* (incalculable) are mentioned. In Jain cosmology, *asankhyeya* is used to represent numbers as large as 10^{140} .

Vedic way of calculating the speed of light

- Sayanacharya gives us the speed of sunlight in $\frac{1}{2}$ nimesa is 2202 yojanas

1 nimesa = $\frac{16}{75}$ seconds i.e., $\frac{1}{2}$ nimesa = $\frac{8}{75}$ seconds

- 1 yojana = 4 kosas; 1 kosa = 2000 dandas; 1 danda = 2 yards;

So, 1 yojana = $4 \times 2000 \times 2 = 16000$ yards;

- We know that 1760 yards = 1 mile i.e., 1 yojana = $\frac{16000}{1760}$ miles = $\frac{100}{11}$ miles

Thus, 2202 yojanas = $2202 \times \frac{100}{11} = \frac{220200}{11}$ miles

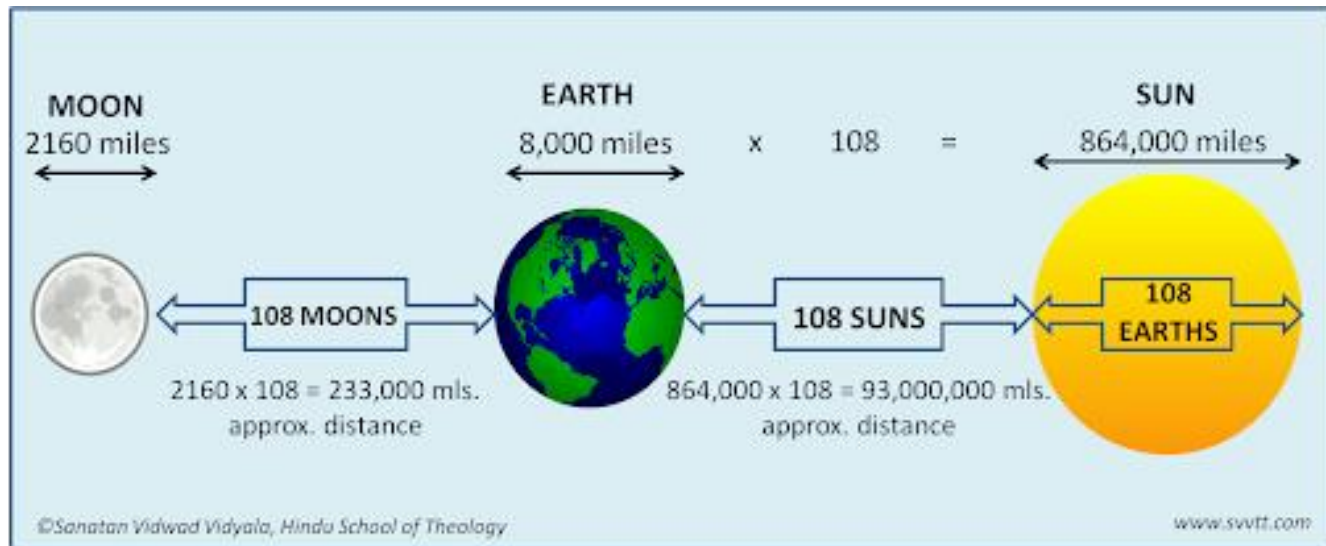
- So, sunlight travels $\frac{220200}{11}$ miles in $\frac{8}{75}$ seconds

The speed of light in 1 second = $(\frac{220200}{11}) \div (\frac{8}{75}) = 1,87,670$ miles

- Actual speed of light is 1,86,282 miles i.e., error is 0.74% only

Symbolic Meaning of Numbers

- Numbers often had cosmological significance
- Three represented the three worlds
- Seven symbolized completeness in many hymns
- Twelve was linked to the solar year
- Thus numbers were both quantitative and symbolic



Baudhayana Sulbasutras (800 BCE)

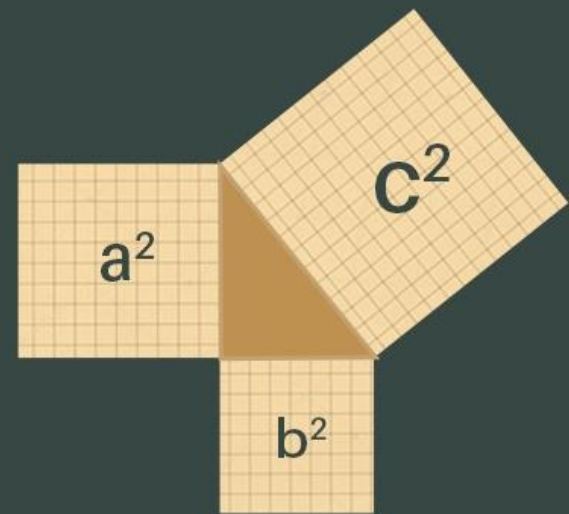
- **Quadratic Equations:** $ax^2 = c$ and $ax^2 + bx = c$ (linear equation in a single unknown)
- **Values of π :** Several approximate values of π found constructing circular shapes
- **Square root of 2 :** $1 + 1/3 + 1/(3 \times 4) - 1/(3 \times 4 \times 34) = 1.41421568$
Modern value of $\sqrt{2} = 1.41421356$

Baudhayana Sulbasutra

दीर्घचतुरश्रस्याक्षण्या रज्जुः पार्श्वमानी तिर्यग्
मानी च यत् पृथग् भूते कुरुतस्तदुभयं करोति ॥

Baudhāyana used a rope as an example in the above shloka/verse, which can be translated as:

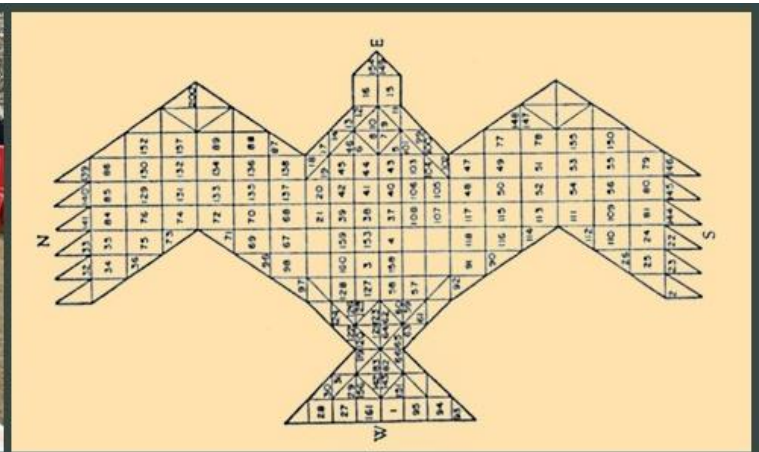
"A rope stretched along the length of the diagonal produces an area which the vertical and horizontal sides makes together"



$$c^2 = a^2 + b^2$$

Apastamba Sulbasutras (600 BCE)

- It has more extensive work (six chapters) than *Baudhanayana Sutra* (three chapters)
- He improved and expanded on the rules given by Bodhayana
- The general linear equation was solved in the *Apastamba Sulbasutra*
- His formula gives a remarkably accurate value of 'square root of 2' correct to five decimal places



The Sulbasutras is like a guide to the Vedas which formulate rules for constructing Vedi (altars). They contain valuable information about the geometry of ancient India, including instructions for constructing sacrificial altars (vedi) and locating sacred fires (agni). This formula is used in constructing Yagna Vedi (altars) in the shape of an eagle with Triangular tiles in a special type of Yagna called Agnicayana (a twelve day Yajna ceremony)

Manava Sulbasutras

Manava Sulbasutras

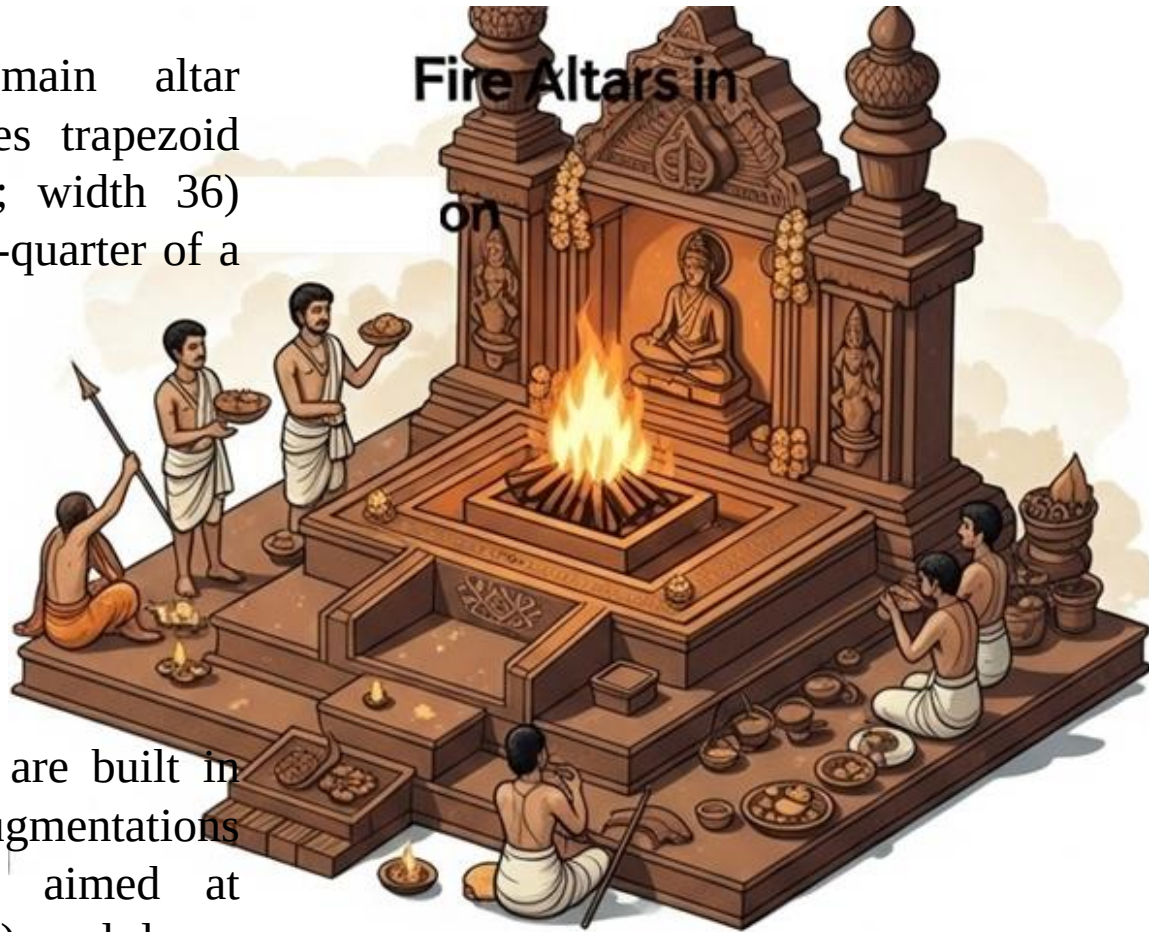
Find the Value of π in Sulba Sutras

1. The Manava Sulbasutras are ancient Indian mathematical texts associated with the construction of Vedic sacrificial altars.
2. These texts provide geometric methods for accurate measurements and intricate geometric constructions.
3. The Manava Sulbasutras introduce the concept of the diagonal of a rectangle and offer approximations for the square root of 2.
4. These mathematical treatises are attributed to the sage Manava or Manu, believed to have lived around 800-500 BCE.



Counting in Ritual Practice

- **Altar Dimensions:** The main altar (mahāvedi) is often an isosceles trapezoid with dimensions (bases 24, 30; width 36) summing to 90, representing one-quarter of a 360-day year



- **Time and Geometry:** Altars are built in layers, with 95 annual augmentations representing a 95-year cycle, aimed at reconciling solar (365/366 days) and lunar (354 days) years
- **Number of Bricks:** A central aspect of Agnicayana is the usage of 10,800 Muhurtas (units of time) in a 1,000-year cycle, which is reflected in the number of bricks used in the construction of the altar

Counting in Ritual Practice

- **Geometric Precision:** The Sulbasutras define methods to calculate irrational numbers like $\sqrt{2}$ (dvīkaraṇī) to transform square altars into circles or other shapes without changing the area
- **Symbolic Numbers:** Altars are designed in specific shapes (falcon, tortoise) using geometric, numeric, and symbolic calculations to represent the universe

Archaeological Evidences



Excavated Vedic symbol (tat rembrding)



Excavated Fire altar
Indter 'hdines

Construction & Geometry of Fire Altars



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Re netial \firewms



Weto stior alor
Firetriss llttrs
(froerittod (Yajna)

Study Tips

Focus Areas

- What to emnehiize study about Fire Altas 1-10 1-10
- E. structure answers, Usie tjn tiam Retywords keywords, reyitatre adid roles.
- Wha lo Am'emrdt. Connect cultural tgnllional significance with social structures.

Rituals & Offerings

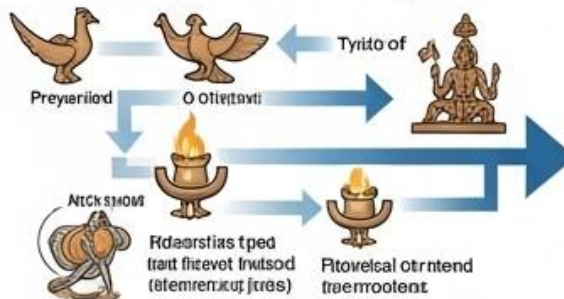


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hymist arm and sides

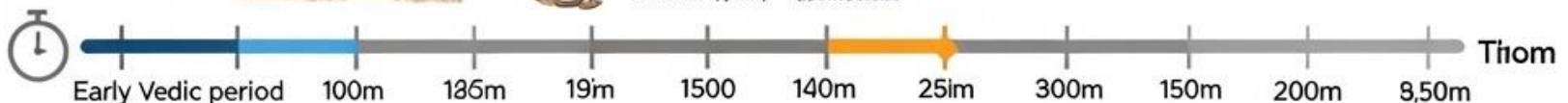
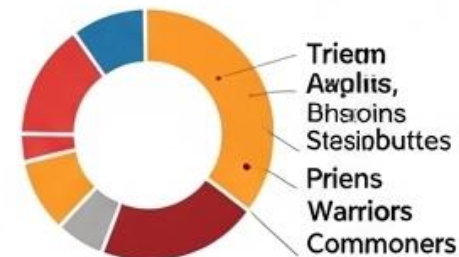


Excavted Fire alts
Bhle Priest

Rituals & Offerings



Societal Impact



Number Sequences

- Sequences of numbers appear in liturgical recitations
- They sometimes increase in regular patterns
- These may reflect early combinatorial thinking
- Ordered sequences helped memory and oral transmission

$$N_n = N_{n-1} + N_{n-2} \text{ for } n > 2, N_0 = N_1 = N_2 = 1$$

- Pattern recognition was important in Vedic culture

Arithmetic Sequence	Geometric Sequence	Triangular Sequence
$x_n = x_1 + (n-1) d$	$x_n = x_1 \times r^{n-1}$	$x_n = (n^2 + n) \div 2$
Cube Sequence	Fibonacci Sequence	Square Sequence
$x_n = n^3$	$a_{n+2} = a_{n+1} + a_n$	$x_n = n^2$

Number Sequences

एका च मे तिस्रश्च मे
पञ्च च मे सप्त च मे
नव च मऽएकादश च
त्रयोदश च मे पञ्चदश च मे
सप्तदश च मे नवदश च मे
ऽएकविंशतिश्च मे त्रयोविंशतिश्च मे
पञ्चविंशतिश्च मे सप्तविंशतिश्च मे
नवविंशतिश्च मऽएकत्रिंशतिश्च

तिस्रश्च मे पञ्च च मे
सप्त च मे नव च
मऽएकादश च मे त्रयोदश च मे
पञ्चदश च मे सप्तदश च मे
नवदश च मऽएकविंशतिश्च मे
त्रयोविंशतिश्च मे पञ्चविंशतिश्च मे
सप्तविंशतिश्च मे नवविंशतिश्च मे
मऽएकत्रिंशतिश्च मे यज्ञेन कल्पताम्॥
— यजुर्वेद १८.२४

*ekā ca me tisraśca me
pañca ca me sapta ca me
nava ca me ekādaśa ca me
trayodaśa ca me pañcadaśa ca me
saptadaśa ca me navadaśa ca me
ekavimśatiśca me trayovimśatiśca me
pañcavimśatiśca me saptavimśatiśca me
navavimśatiśca ma ekatrimśaśca
ma ekatrimśaśca me yajñena kalpatām ॥*

*tisraśca me pañca ca me
sapta ca me nava ca me
ekādaśa ca me trayodaśa ca me
pañcadaśa ca me saptadaśa ca me
navadaśa ca me ekavimśatiśca me
trayovimśatiśca me pañcavimśatiśca me
saptavimśatiśca me navavimśatiśca me*

— Yajurveda, 18.24

- The sequence of pair of numbers given here is (1,3), (3,5), (5, 7), (7, 9), (9, 11), (11, 13)(29, 31). These are pair of odd numbers.
- Several pattern can arise by their pairwise sum, difference and product
- Sum : 4, 8, 12, 16, 20, 24,...60

- Product : 3, 15, 35, 63, 99, ...899
- Difference of the product series : 12, 20, 28, 36,116

Number Sequences

चतस्रश्च मेऽष्टौ च मे

द्वादश च मे षोडश च मे

विंशतिश्च मे चतुर्विंशतिश्च मे

ऽष्टाविंशतिश्च मे द्वात्रिंशतिश्च मे

षट्त्रिंशतिच्च मे चत्वारिंशतिच्च मे

चत्वारिंशतिच्च मे चतुश्चत्वारिंशतिच्च मे

चतुश्चत्वारिंशतिच्च मेऽष्टचत्वारिंशतिच्च मे यज्ञेन कल्पताम्॥२५॥

ऽष्टौ च मे द्वादश च मे

षोडश च मे विंशतिश्च मे

चतुर्विंशतिश्च मे ऽष्टाविंशतिश्च मे

द्वात्रिंशतिच्च मे षट्त्रिंशतिच्च मे

catasraśca me aṣṭau ca me

dvādaśa ca me ṣoḍaśa ca me

viṁśatiśca me caturviṁśatiśca me

aṣṭāviṁśatiśca me dvātriṁśatiśca me

ṣaṭtriṁśaticca me catvāriṁśaticca me

catuṣcatvāriṁśaticca me aṣṭcatvāriṁśaticca me

aṣṭau ca me dvādaśa ca me

śoḍaśa ca me viṁśatiśca me

caturviṁśatiśca me aṣṭāviṁśatiśca me

dvātriṁśaticca me ṣaṭtriṁśaticca me

catvāriṁśaticca me catuṣcatvāriṁśaticca me

catuṣcatvāriṁśaticca me yajñena kalpatām

— Yajurveda, 18.25

- The sequence of pair of numbers given here is (4, 8), (8, 12), (12, 16), (16, 20)

Infinity in Vedic Thought

- The idea of the infinite appears in philosophical passages
- The famous phrase ‘purnam adah purnam idam’ suggests completeness
- Numbers were seen as extendable without bound
- This philosophical infinity influenced later mathematics
- It laid groundwork for comfort with large numbers

om pūrṇamadaḥ pūrṇamidaṁ pūrṇātpūrṇamudacyate
pūrṇasya pūrṇamādāya pūrṇamevāvaśiṣyate

ॐ पूर्णमदः पूर्णमिदं पूर्णात्पूर्णमुदच्यते ।
पूर्णस्य पूर्णमादाय पूर्णमेवावशिष्यते ॥

That is full, this also is full.
This Fullness came from that Fullness.
Though this Fullness came from that Fullness,
That Fullness remains forever full.

Concept of Zero (Early Notions)

- India invented zero (*śūnya*) both as a symbol and as a number
- The earliest known use of a zero like symbol in India is found in Pingala's *Chandah-sutra* (200 BCE), appeared as a metrical context
- In Bakhshali manuscript (200 CE), zero appeared as a number denoted by small dot

Babylonians



Mayans



Chinese



Hindus



- Brahmagupta (598 – 668 CE), in his *Brahmasphutasiddhanta* treated zero as a number and gave rules for using it in calculations
- The use of zero completed decimal place value system which required only ten symbols (0 – 9)

Arithmetic in the Vedic Period

- Arithmetic developed from practical needs, basic operations like addition and subtraction were known
- Positive (dhana) and negative (rna) numbers mentioned in Vasistha Siddhanta, Brahmasphutasiddhanta (Brahmgagupta)
- Division concepts were implied in sharing rituals. The first instance of the operation of division is from the Satapatha Brahmana
- Prajapati is the year with 720 days and nights at one level and, at another, the ritual altar having the same number invoked in the Taittirīya Samhitā
- Bakshali manuscript has mentioned negative numbers as well as bindu (zero)

Cyclic Quadrilateral

- Three non-collinear points in a plane lie on a unique circle i.e., all the three vertices of a triangle lie on a circle i.e., every triangle is cyclic
- A cyclic quadrilateral whose all four vertices lie on the same circle. However all quadrilateral is not cyclic.
e.g., every rectangle is cyclic quadrilateral but parallelogram (not rectangle) and trapezium with unequal non-parallel sides is not cyclic
- Brahmagupta (628 CE) gave a special formula to calculate the area of a cyclic quadrilateral and also described some important properties of its diagonals

Cyclic Quadrilateral

- **Theorem** : If a, b, c, d are the lengths of the sides of a cyclic quadrilateral and x, y are its diagonals, then

$$x = \sqrt{\frac{(ab+cd)(ac+bd)}{(ad+bc)}} \quad y = \sqrt{\frac{(ad+bc)(ac+bd)}{(ab+cd)}} \quad \text{i.e., } xy = ac + bd$$

- **Area of cyclic quadrilateral:**

$$A = \sqrt{s(s-a)(s-b)(s-c)(s-d)}$$

Where s is the semi perimeter, $s = \frac{a+b+c+d}{2}$

- **Circumradius of a cyclic quadrilateral**

$$R = \frac{1}{4} \sqrt{\frac{(ab+cd)(ac+bd)(ad+bc)}{s(s-a)(s-b)(s-c)(s-d)}} = \frac{abcd}{4K} \sqrt{\frac{(ab+cd)(ac+bd)(ad+bc)}{a^2b^2c^2d^2}}$$

Where K is the area of cyclic quadrilateral

Fractions and Parts (Bhaga/Amsa)

- Texts (e.g., *Maitrayani Samhita*, *Yajurveda*) describe parts of offerings and measures [Ardha ($1/2$), Pada ($1/4$), Sapha ($1/8$), Kustha ($1/12$), Kala ($1/16$)], fractions were used in ritual measurements
- This implies familiarity with fractional quantities (Yajurveda mentions series of fractions like $1/2$, 1, $1\frac{1}{2}$, 2, $2\frac{1}{2}$, 3, $3\frac{1}{2}$, 4)
- Ritual accuracy required division of materials (the conversion from circle to square required to multiply by $13/15$)
- Though not formalized, fraction thinking was present and later mathematics would systematize fractions
- However Indian mathematics never adopted the decimal representation of fractions throughout the course of its history

Combinatorial Ideas

- Prosody and hymn structures involved counting patterns, meters depended on arrangements of syllables, this led to early combinatorial reasoning
- Later scholars like Pingala (300 – 200 BCE) developed this further in Chandahsutra
- Thus poetry influenced mathematical development
- Combinatorial methods in Indian music : Pratyayas in Sangitaratnakara of Sarngadeva
- A unified mathematical treatment on the theory of binomial coefficients by Narayana

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n-1} + \binom{n}{n} = 2^n$$

Chandaḥśāstra

- The [Chandaḥśāstra](#) (or Chandahsutra), composed by Acharya Pingala around the 3rd century BCE
- It is the foundational Sanskrit text on prosody (the study of poetic meters) and an early treatise on combinatorics and mathematics
- यमाता-राजभान-सलगाम् (**Yamātārājabhānasalagam**) is a famous Sanskrit mnemonic used in prosody (*Chandas*) to remember the structure of the eight **Ganas** (syllable groups) of binary numbers each of word length 3
- Here the syllables in Sanskrit occur in two durations : short which is called laghu (light) and long or guru (heavy, of twice the duration of the short)

Combinatorics

Gana	Syllables	Pattern I = Laghu (Short/Light), , S = Guru (Long/Heavy)	Description
Ya (यगण)	Ya-mā-tā	I S S	One light, two heavy
Mā (मगण)	Mā-tā-rā	S S S	Three heavy
Tā (तगण)	Tā-rā-ja	S S I	Two heavy, one light
Rā (रगण)	Rā-ja-bhā	S I S	Heavy, light, heavy
Ja (जगण)	Ja-bhā-na	I S I	Light, heavy, light
Bhā (भगण)	Bhā-na-sa	S I I	One heavy, two light
Na (नगण)	Na-sa-la	I I I	Three light
Sa (सगण)	Sa-la-gaṃ	I I S	Two light, one heavy

Powers of Ten

- Lists of powers of ten appear in later Vedic texts
- Names existed for numbers up to extremely high powers
- This demonstrates systematic numerical scaling
- It reflects early place-value awareness
- Such lists are unique in ancient cultures

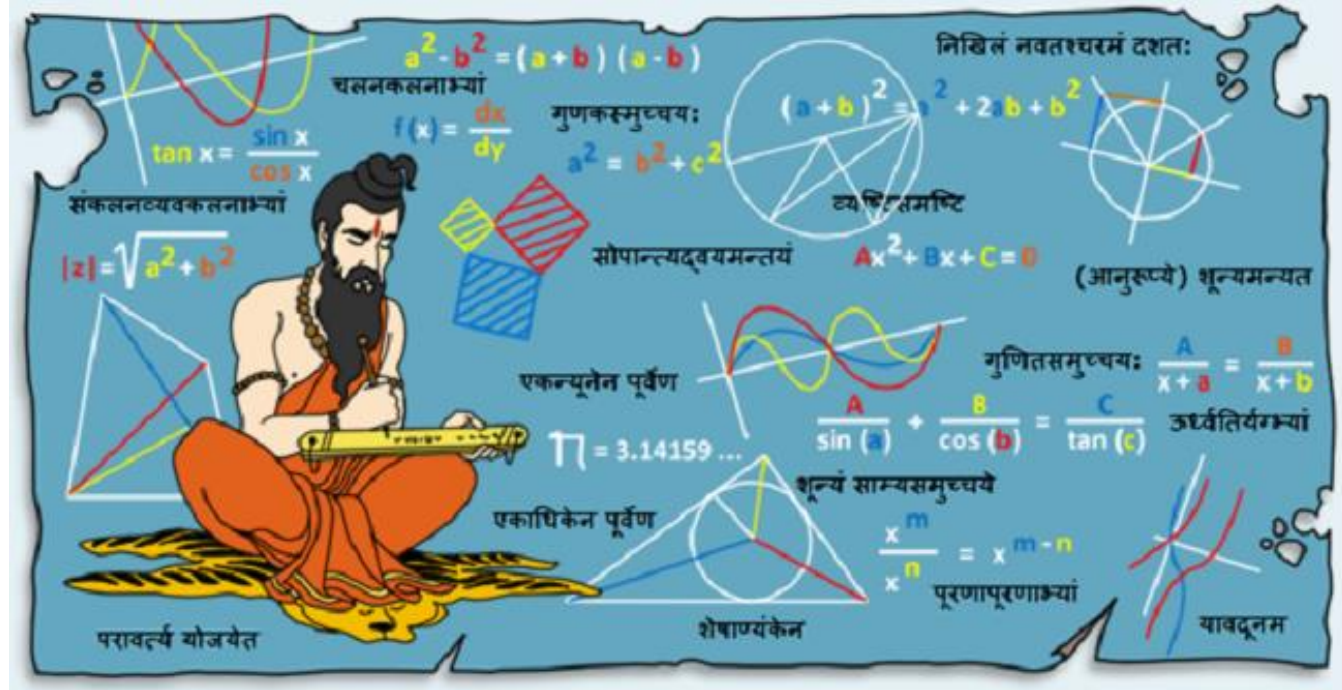
Powers of Ten in Ancient Indian Mathematics

<i>daśa</i>	<i>daśa</i>	Ten	10^1
<i>daśa daśa</i>	<i>śata</i>	Hundred	10^2
<i>daśa śata</i>	<i>sahasra</i>	Thousand	10^3
<i>daśa sahasra</i>	<i>ayuta</i>	Ten Thousand	10^4
<i>daśa ayuta</i>	<i>niyuta</i>	Hundred Thousand	10^5
<i>daśa niyuta</i>	<i>prayuta</i>	Million	10^6
<i>daśa prayuta</i>	<i>arbuda</i>	Ten Million	10^7
<i>daśa arbuda</i>	<i>nyarbuda</i>	Hundred Million	10^8
<i>daśa nyarbuda</i>	<i>samudra</i>	Thousand Million	10^9
<i>daśa samudra</i>	<i>madhya</i>	Ten Thousand Million	10^{10}
<i>daśa madhya</i>	<i>anta</i>	Hundred Million Thousand	10^{11}
<i>daśa anta</i>	<i>parārdha</i>	Billion	10^{12}

Reference :
**Vajasaneyi
Samhita** of
Shukla
Yajurveda

Cultural Importance of Enumeration

- Enumeration was not merely practical but sacred
- Counting was tied to order and harmony
- Numbers represented structure in the universe
- This gave mathematics a cultural prestige
- It encouraged preservation of numerical knowledge



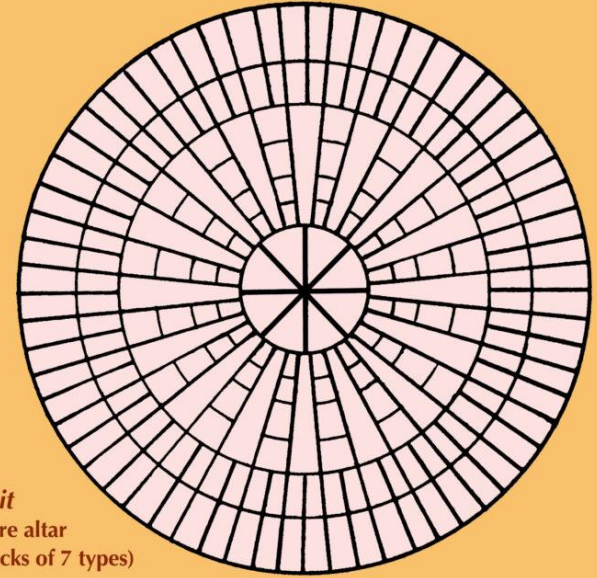
Languages of Indian Mathematical Knowledge

- Most of the ancient Indian mathematics was written in the *Sanskrit* language and presented in the form of verses to make it easy to remember and pass on through oral tradition
- Some Jain texts, like the *Sūryaprajñapti* and *Anuyogadvārasūtra*, which include mathematical ideas, were written in Prakrit
- While some Buddhist texts that talk about large numbers and cosmic time cycles were written in *Pali*

Sources of Indian Mathematical Knowledge

- Texts (Shastras and Sutras) : Śulbasūtras, Chandahśāstra, Āryabhatīya, Sūryasiddhānta, Brāhmasphutasiddhānta, Līlāvatī, and Bījaganita
- Commentaries and teaching traditions
- Manuscripts and Inscriptions
- Oral traditions and memory techniques : *Katapayadi* and *Bhut Sankhya* number system
- Foreign accounts : Al-Biruni and Hsüan-Tsang

Appendices

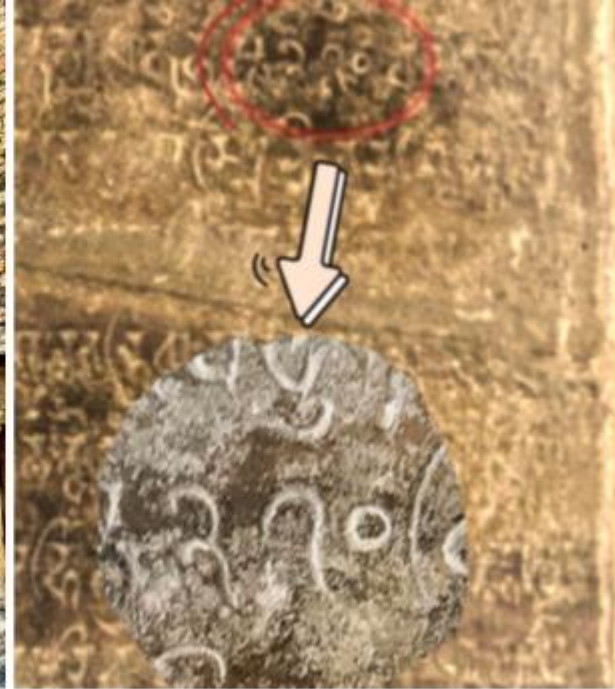


The *rathacakracit*
or chariot-wheel fire altar
(first layer, 200 bricks of 7 types)



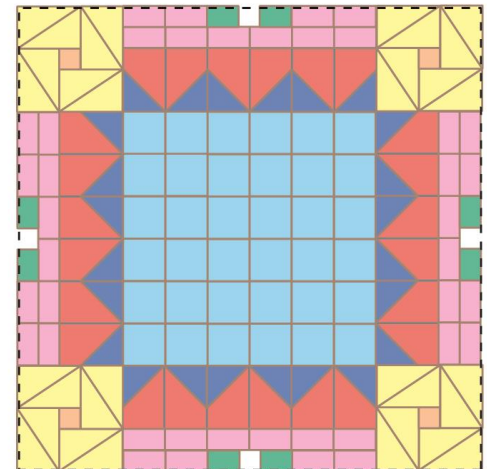
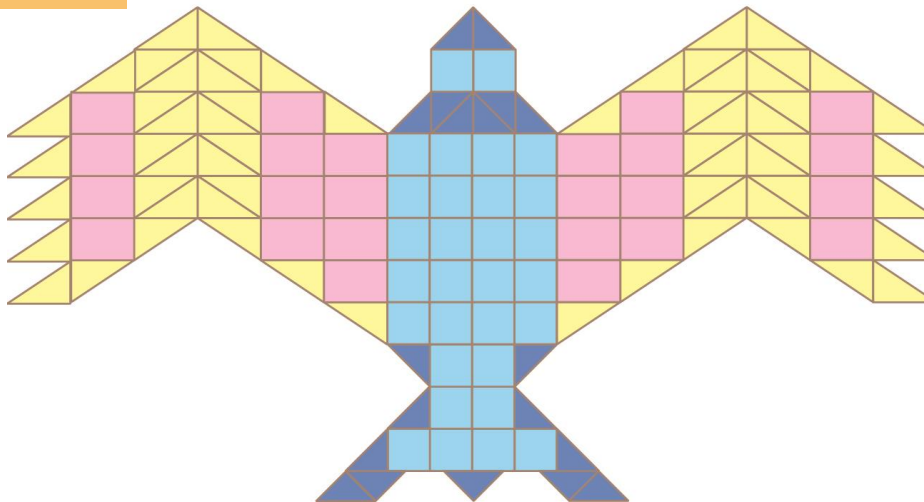
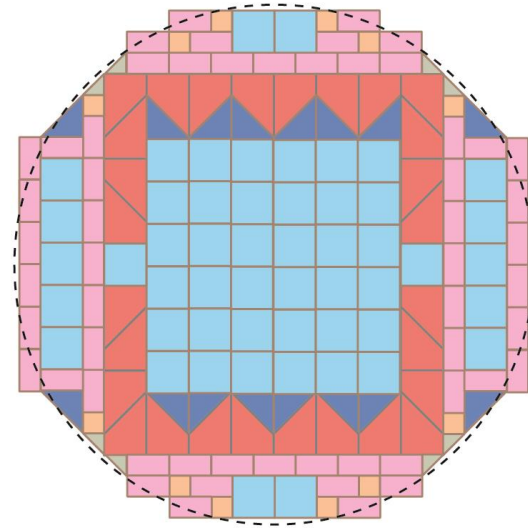
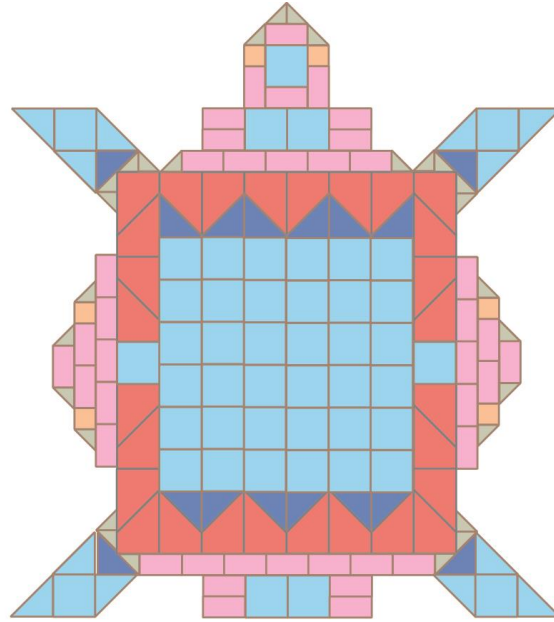
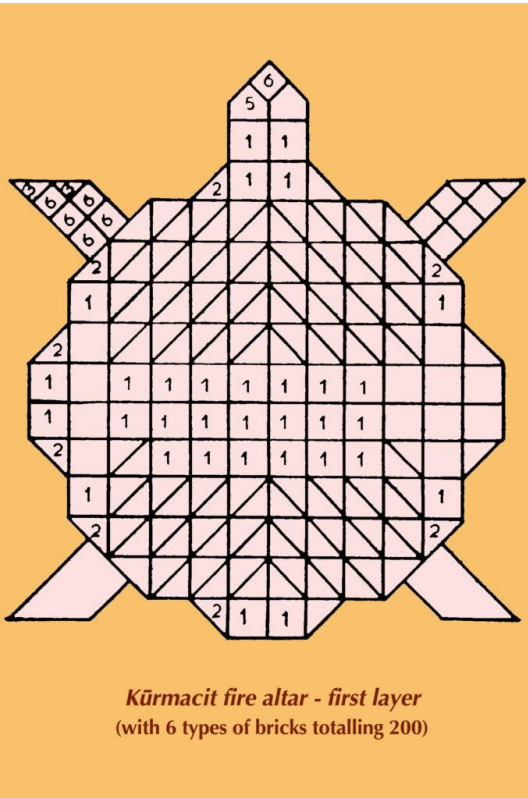
Bakhshali Manuscript

Appendices



Chaturbhuj Temple

Altar Variations



Name	Sanskrit (Devanagari)	Number	Power of 10
Eka	एक	1	10^0
Daśa	दश	10	10^1
Śata	शत	100	10^2
Sahasra	सहस्र	1,000	10^3
Ayuta	अयु त	10,000	10^4
Lakṣa	लक्ष	100,000	10^5
Prayuta	प्रयु त	1,000,000	10^6
Koṭi	कोटि	10,000,000	10^7
Arbuda	अर्बुद	100,000,000	10^8
Abja	अब्ज	1,000,000,000	10^9
Kharva	खर्व	10,000,000,000	10^{10}
Nikharva	निखर्व	100,000,000,000	10^{11}
Mahāsaroja	महासरोज	1,000,000,000,000	10^{12}
Śanku	शङ्कु	10,000,000,000,000	10^{13}
Saritāpati	सरितापति	100,000,000,000,000	10^{14}
Antya	अन्त्य	1,000,000,000,000,000	10^{15}
Madhya	मध्य	10,000,000,000,000,000	10^{16}
Parārdha	परार्ध	100,000,000,000,000,000	10^{17}

Reference : Sridhara (750 CE)